
COMMUNITY
SUMMIT
Microsoft Fabric

The largest independent Microsoft AI & Business Applications User Conference on the planet.

COMMUNITY SUMMIT

Dataverse for the Fabric Practitioner: What to Know Before You Build



@ 2026 Dynamic Communities

HI, I'M JASON

THE DATA SHEPHERD

Microsoft® Most Valuable Professional

Honored to be recognized by Microsoft.



I HERD THE DATA

- User Group Organizer
- Nashville, Tennessee
- Fabric + Power BI
- Teaching + Speaking







MY FAVORITE MODEL

TARDIS

TOOLS & ALLIES

Power BI | Microsoft Fabric | DuckDB | Python | SQL | VS Code

Our Journey

Still Under Construction

A real implementation. Lessons as we build.

BEFORE YOU BUILD

The Build Plan

05 · VALIDATE	Pass Inspection
04 · MODEL	Build for Analytics
03 · CONNECT	Connect the Utilities
02 · INSPECT	Inspect the Blueprint
01 · UNDERSTAND	Understand the Site

THE PLAN

Five phases. Nothing laid yet.

BEFORE YOU BUILD

The Build Plan

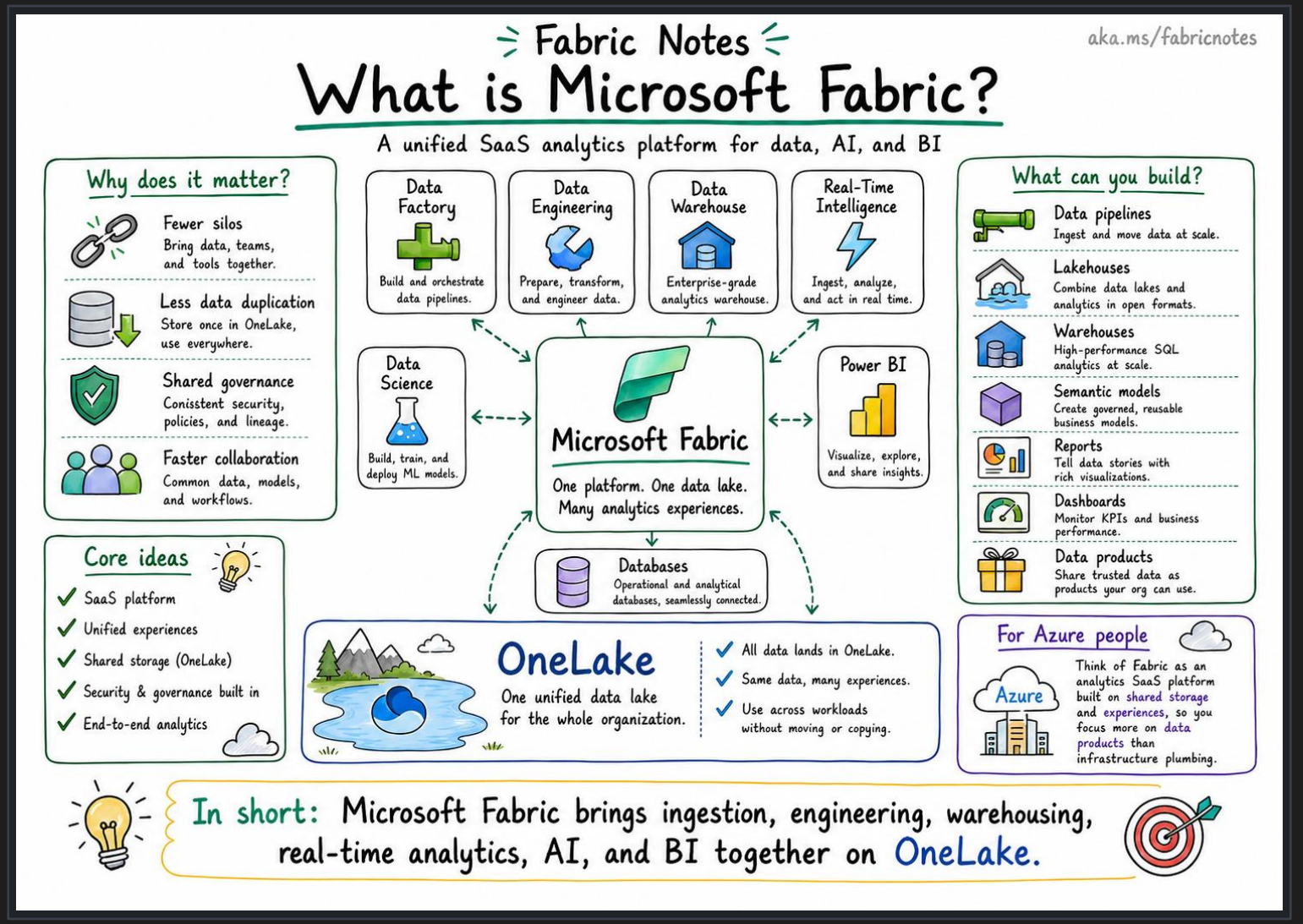
01 · UNDERSTAND

Understand the Site

NOW LAYING

01 · Understand the Site — You didn’t design this schema. An application did.

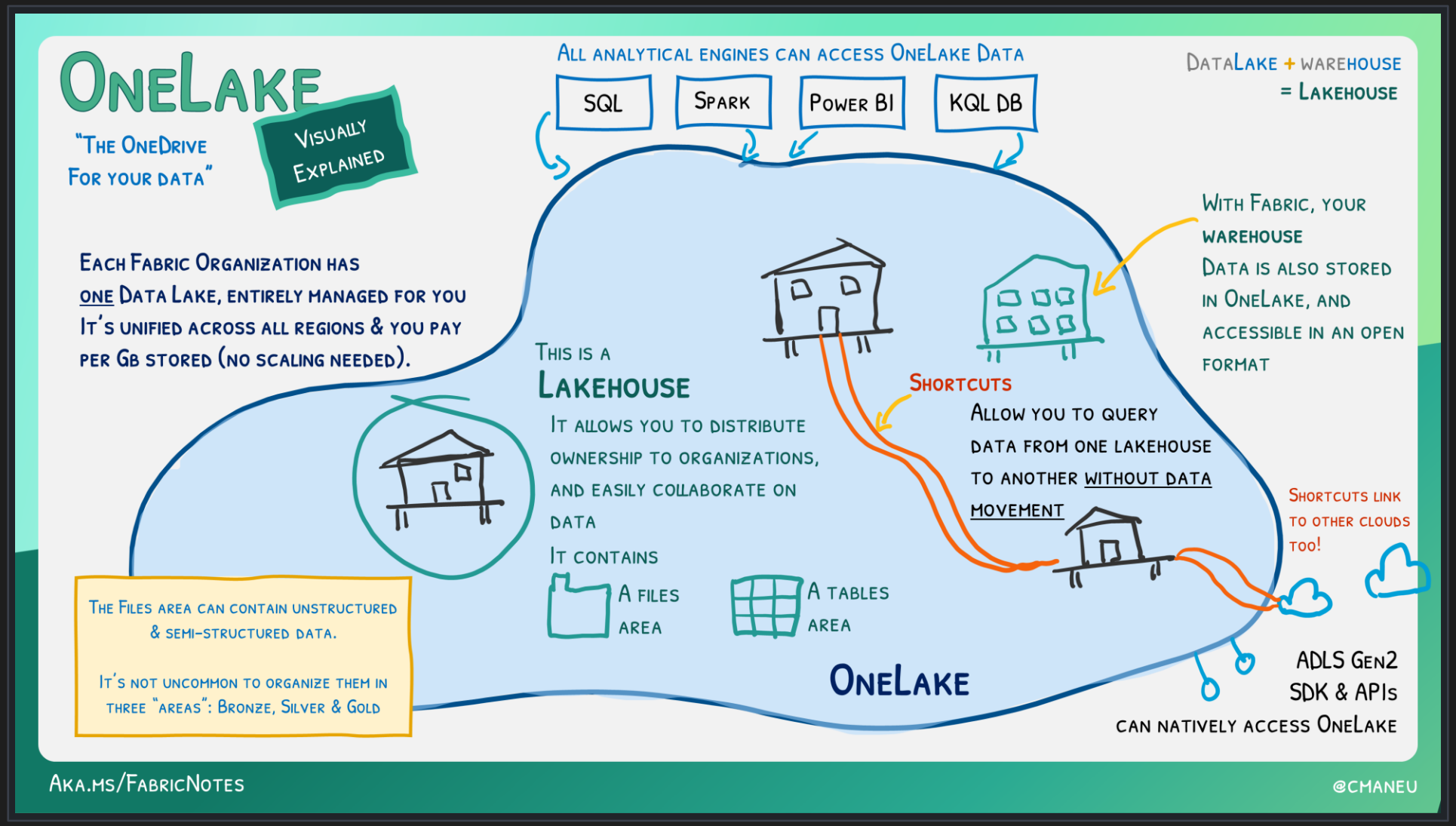
What is Microsoft Fabric?



One unified SaaS platform for data, AI and BI — every workload on a single data lake.

Source: Fabric Notes · aka.ms/fabricnotes

OneLake, visually explained



One data lake per organization — shortcuts let every engine query data in place, with no copies.

Source: Fabric Notes · aka.ms/fabricnotes

01 · UNDERSTAND THE SITE

What is Dataverse?

Microsoft’s managed data platform for business applications

BUSINESS DATA

Tables and relationships hold business records. Metadata describes their structure and meaning.

APPLICATIONS

Dynamics 365 Sales, Customer Service and Field Service use Dataverse. Model-driven Power Apps use it too.

PLATFORM SERVICES

Security controls access. Business rules, plug-ins and automation support the application’s processes.

Business Central and finance and operations apps have their own data stores. They can integrate with Dataverse.

Understand the application behind the data

Tables support the application's business process. Solutions introduce and extend the schema.

01

APPLICATION

Start with the workload and the business process it supports.

02

SOLUTION / PUBLISHER

Find the solution and publisher behind the table or customizations.

03

TABLE

Inspect the logical name and ask the owning team what its rows mean.

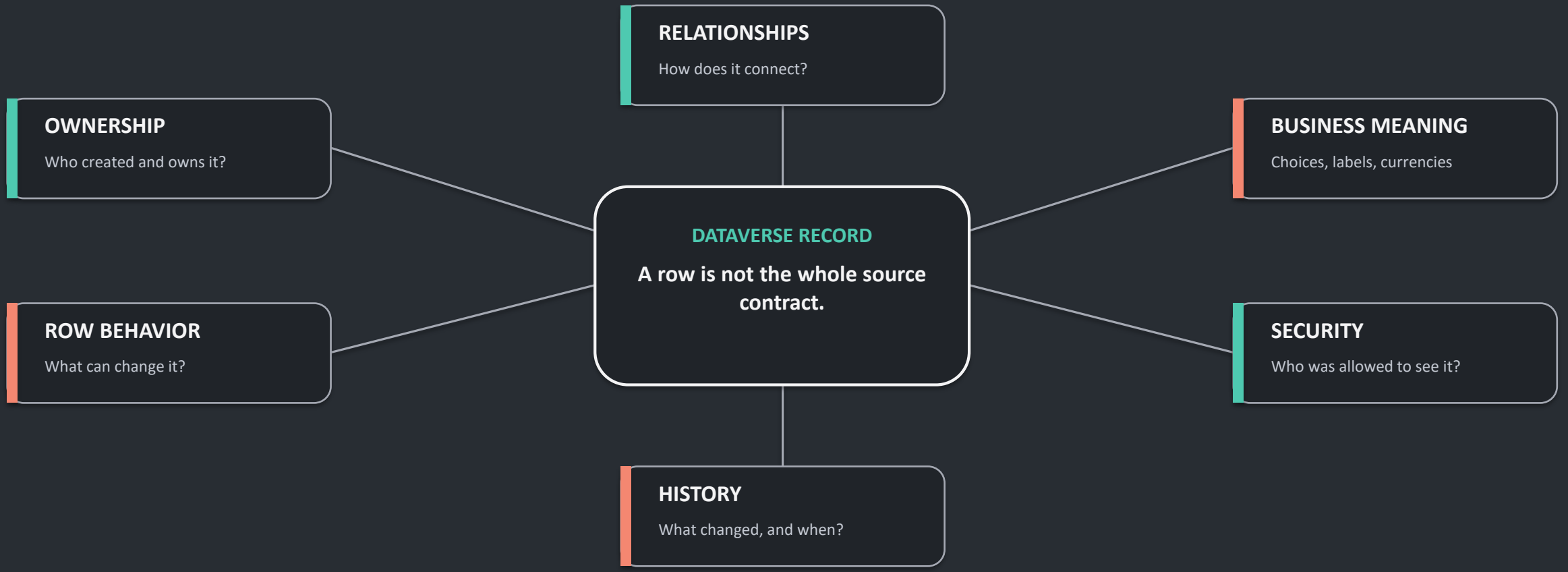
EXAMPLE: ACCOUNT

Dynamics 365 Sales uses account. Your solutions can add columns and behavior.

01 - UNDERSTAND THE SITE

The rows are only part of the picture.

Moving the rows doesn’t automatically preserve their meaning, rules or access controls.



Before you model the data, inspect the application context that gives it meaning.

BEFORE YOU BUILD

The Build Plan

02 · INSPECT	**Inspect the Blueprint**
01 · UNDERSTAND	Understand the Site

NOW LAYING

02 · Inspect the Blueprint — What that actually looks like when you go to query it.

Display names, logical names and entity sets

Use metadata to identify the name your query or connection expects.

Display name	Logical name	Web API entity set

Confirm names in metadata. Do not derive an entity set by pluralizing a name.

Keep a metadata reference for the tables you use; refresh it when the schema changes.

02 · INSPECT THE BLUEPRINT

Column types need business interpretation

Inspect the Dataverse definition and the values exposed by your chosen route.

CHOICE

A numeric code represents a labeled value.

Inspect local or global choices. Map codes to labels in the language your report needs.

LOOKUP

A reference identifies a row in another table.

Inspect the target table or tables before joining. Customer and Owner can refer to different table types.

CURRENCY

Money has currency and precision context.

Check the currency, exchange rate and base amount before aggregating.

DATE AND TIME

Date/time behavior controls time zone conversion.

Check User Local, Date Only or Time Zone Independent before transforming it.

02 · INSPECT THE BLUEPRINT

Row identity and business uniqueness

PRIMARY KEY

accountid identifies a row. Its type is GUID.

ALTERNATE KEYS

Business columns can enforce uniqueness.

ANALYTICAL GRAIN

This reporting view has one row per Account.

- PRIMARY KEY: Join on IDs. The primary name is a display value and can repeat.
- ALTERNATE KEYS: The Keys page lists these. Check Active status, nulls and duplicates.
- GRAIN: Joining all Contacts can multiply Account rows. Check rows and distinct IDs.

DATAVERSE · COMMUNITY TOOLS

What is XrmToolBox?

A free, community-built Windows application
that connects to Microsoft Dataverse.

FREE

COMMUNITY-BUILT

EXTENSIBLE

Discover and install tools from the Tool Library.

Built by the Dataverse community for makers, administrators and developers.

DATAVERSE · COMMUNITY TOOLS

A community toolbelt for Dataverse

Different tools for the jobs you encounter in a Dataverse environment.

Understand the schema	Metadata browsers and documentation tools
Explore the data	Query builders and SQL tools
Manage an environment	Administration and security tools
Configure applications	Customization and solution tools
Investigate a problem	Diagnostics and troubleshooting tools

For a Fabric practitioner: another way to understand the source before building downstream.

Understand Account columns before joining

Identify the related table and business role before designing your joins.

ACCOUNT COLUMN	REFERENCE OR VALUE	WHAT TO INSPECT
accountid	The row’s Dataverse identity	Primary key and related-table references
primarycontactid	A link to a contact	Contact target and the primary-contact role
ownerid	A user or team owner	User or team target; ownership is only part of access control.
statecode / statuscode	Lifecycle codes	Choice labels and valid state/status combinations

Confirm relationship targets and cardinality in metadata. Validate the rows you actually receive.

Stop trying to make fetch happen.

FetchXML is Dataverse's XML-based query language.
FetchXML Builder lets you select columns, filters and relationships visually.

```
<fetch top="10">
  <entity name="account">
    <attribute name="accountid" />
    <attribute name="name" />
    <attribute name="createdon" />
  </entity>
</fetch>
```

account

The table's logical name

attribute

A column to return

top="10"

Return up to ten records

Choose the columns you need. Build visually, then inspect and run the query.

02 · INSPECT THE BLUEPRINT

Dataverse connection and metadata toolkit

CONNECTION EXAMPLES

Compare Dataverse / TDS queries with a focused Web API + FetchXML query.

METADATA INSPECTION

Explore table names, primary IDs, column types and lookup targets.

REUSABLE SETUP

Change environment parameters and select tables in Power Query.

- COMPARE: Full Account data and a focused FetchXML sample show different query shapes.
- INSPECT: Dataverse metadata supplies business meaning. SQL metadata shows exposed types.
- REUSE: Follow Connections and Setup, sign in to your environment, then refresh selected queries.

DEMO 1

Inspect the Blueprint

Measure twice. Query once.

XrmToolBox / SQL 4 CDS / Excel

BEFORE YOU BUILD

The Build Plan

03 · CONNECT	**Connect the Utilities**
02 · INSPECT	Inspect the Blueprint
01 · UNDERSTAND	Understand the Site

NOW LAYING

03 · Connect the Utilities — Three routes, different tradeoffs, and new Fabric Link choices.

03 · CONNECT THE UTILITIES

Query the source or replicate for analytics

Choose the route based on workload, freshness and access requirements.

ROUTE	WHAT IT IS	FRESHNESS	ACCESS EVALUATED THROUGH
TDS endpoint	Read-only SQL over Dataverse	At query time	**Query identity**
Dataverse connector	Power BI via TDS; Import or DirectQuery	Import: last refresh DirectQuery: query time	**Refresh/query identity**
Web API (OData)	REST API; FetchXML queries supported	At query time	**Request identity**
Link to Fabric	Replica exposed through OneLake shortcuts	Asynchronous	**Fabric permissions**
Azure Synapse Link	Exports Dataverse data to your Azure data lake	Asynchronous	**Azure destination permissions**

Import stores a refreshed copy. DirectQuery queries the source. Link options provide asynchronous data for analytics.

Where the data lands

The storage owner changes what you can do after the link is created.

AZURE SYNAPSE LINK

Dataverse exports to your ADLS Gen2 account

You own the storage and choose the downstream tools.

FABRIC LINK

Delta replica in Dataverse-managed storage

OneLake shortcuts expose it in a read-only Lakehouse.

THE DECISION

Do you need storage you control, or a managed path into Fabric?

03 · CONNECT THE UTILITIES

Synapse Link or Fabric Link?

Choose the storage and operating model before comparing setup screens.

FABRIC LINK	AZURE SYNAPSE LINK

How Dataverse data reaches Fabric

Fabric provides the analytics platform. OneLake shortcuts expose the linked data.

01 DATAVERSE	Business application records
02 MANAGED DELTA REPLICA	Dataverse-managed storage; maintained by Link to Fabric
03 ONELAKE SHORTCUTS	Expose the replica as tables in the linked lakehouse
04 SQL ANALYTICS ENDPOINT	Query those tables with SQL and build reporting views

The linked lakehouse exposes managed data through shortcuts.
It does not create another copy of each table in your workspace.

03 · CONNECT THE UTILITIES

Replication needs its own access design

SOURCE QUERIES

Dataverse evaluates the querying identity’s permissions.

Import stores data from its refresh identity.

Report viewers need a model access design.

LINK TO FABRIC

The connection identity authorizes the link to Dataverse.

Fabric permissions govern access to the replicated data.

Dataverse row and column security do not transfer automatically.

Define workspace access and model security before sharing the replicated data.

03 · CONNECT THE UTILITIES

Choose a connection identity

Workspace identity is Microsoft's recommended option for Link to Fabric.

ORGANIZATIONAL ACCOUNT

Uses a person's sign-in. Consider ownership continuity and credential lifecycle.

SERVICE PRINCIPAL

Uses an application identity. Configure its Dataverse application user and credential lifecycle.

WORKSPACE IDENTITY

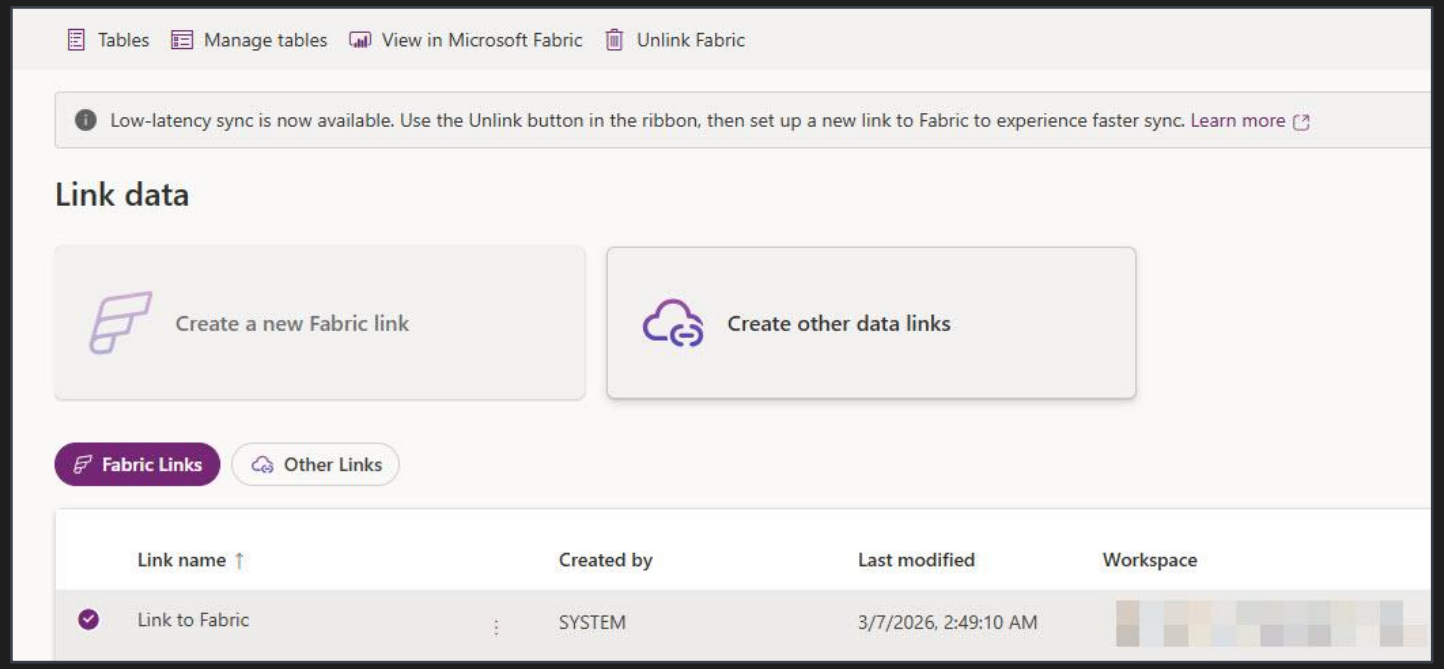
Fabric-managed identity. Add it as a Dataverse application user. No secret for you to manage.

Create the workspace identity in Fabric, then onboard it in Dataverse with the required role.

03 · CONNECT THE UTILITIES

Low-latency sync in Link to Fabric

Also called "Fast Fabric." A newer sync engine within the same connection route.



WHAT CHANGED

Writes directly to Delta Parquet, removing the intermediate CSV step. Check the profile's Low-latency mode flag.

EXISTING LINKS

Older profiles retain the previous engine. Moving requires unlinking and relinking, which removes the linked lakehouse and triggers a full initial sync. Review and repair downstream dependencies.

Review dependencies and format changes before relinking. Measure freshness in your workload.

GA announced June 2026. Profile availability and behavior must be verified in your environment.

Change Data Feed in Fabric Link

Optional change history for downstream incremental processing.

WHAT IT CAPTURES

Insert, update and delete events
Updates can include before / after values
Approximately 24 hours of changes
Takes effect on the next table data change; earlier changes are not captured

WHAT TO CONSIDER

Additional memory and compute work
Can increase sync latency
Consider the latency tradeoff before enabling
Reading current table data does not require CDF

Administrator setting: Power Platform admin center

Environment settings / Product / Features / Microsoft Fabric

SQL endpoint metadata refresh is a separate operation.

Microsoft Learn: Change Data Feed for Fabric Link

An existing Fabric Link blocks a second link



THIS ENVIRONMENT

The current UI allows one Fabric Link.

“Coming soon” is a UI message, not a confirmed release date.

Microsoft announcement: Link to Fabric UX refresh — planned multiple links

03 · CONNECT THE UTILITIES

Freshness to the SQL endpoint has two stages

A change can reach OneLake before the SQL analytics endpoint shows it.

01 DATAVERSE TO ONELAKE

Link to Fabric low-latency sync

Writes source changes to Delta tables exposed through OneLake shortcuts.

02 ONELAKE TO SQL ENDPOINT

New metadata sync (preview)

Makes Lakehouse changes available to queries through the SQL endpoint.

Enable under Workspace settings > Warehouse settings. During preview, it applies to new endpoints only.

Validate freshness where your report queries the data.

SQL analytics endpoint refresh

Lakehouse preview can show rows before SQL discovers the tables.

SYMPTOM

Rows in Lakehouse preview
0 tables in SQL / SSMS

FIRST CHECK

Refresh endpoint metadata
Retry the SQL query

IN THE FABRIC BROWSER

1. Open the SQL analytics endpoint
2. Select Refresh in the Explorer toolbar
3. Retry the query

API / notebook refresh is an automation option.

Refresh checks SQL metadata against lakehouse data; it does not force Dataverse sync or repair every endpoint failure.

Microsoft Learn: SQL analytics endpoint metadata sync

DEMO 2

Connect the Utilities

Let’s see where these pipes go.

Power Apps / Fabric / Power BI

BEFORE YOU BUILD

The Build Plan

04 · MODEL	**Build for Analytics**
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NOW LAYING

04 · Build for Analytics — Patterns that survive contact with an application-owned schema.

04 · BUILD FOR ANALYTICS

Build in a layer you control

Keep the linked replica separate from transformations and business definitions.

LINKED SOURCE

Treat the Dataverse replica as read-only. Use it as the source for analysis or downstream transformations.

SHARED DEFINITIONS

Use SQL views for reusable query logic. Persist transformed tables in your own lakehouse when the workload needs them.

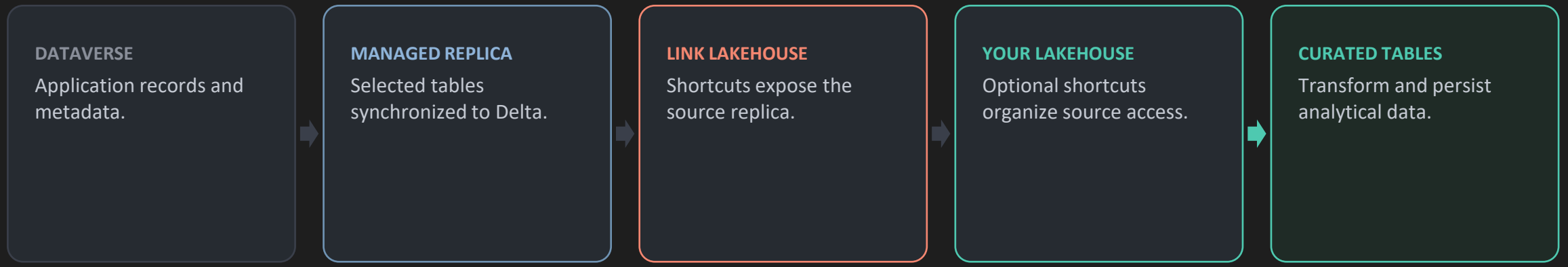
SEMANTIC MODEL

Define relationships, measures and access rules for the report audience. Choose the storage mode deliberately.

Persisted curation is an option when business rules, history or reuse require it.

04 · BUILD FOR ANALYTICS

A Fabric pattern for Dataverse analytics



- Shortcuts reference existing data without an extra ingestion copy. Curated tables use storage you control.
- A schema-enabled lakehouse can organize supported shortcut sources. Validate the shortcut target and access in your environment.
- Build facts, dimensions and a semantic model from the curated data. Plan source dependencies when relinking.

04 · BUILD FOR ANALYTICS

An Account model starts with grain

One Account row, its primary Contact and explicit lifecycle labels.

ACCOUNT GRAIN

One row per Account. Keep accountid as source identity. Names can change or repeat.

PRIMARY CONTACT

Match primarycontactid to contactid. Joining every related Contact can multiply Account rows.

LIFECYCLE

Keep state and status codes. Resolve Status Reason using both State and Status, in the chosen language.

REPORTING RESULT

Count distinct accountid. Filter statecode = 0 for active Accounts. Current state alone cannot show history.

04 · BUILD FOR ANALYTICS

Turn Dataverse values into reporting attributes

Reuse the same label and lookup rules across analytical outputs.

SOURCE VALUE	ANALYTICAL ATTRIBUTE	TRANSFORMATION
Choice code	Business label in the chosen language	Resolve local or global choice metadata; filter by language
Lookup identifier	Account, Contact, User or Team attributes	Resolve target type, then join on its source key
Date/time value	Date and time for the reporting use case	Respect User local, Date only, or Time zone independent behavior

Keep source identifiers and codes alongside curated attributes. Use the semantic model for shared measures and reporting relationships.

Metadata gives linked data its meaning

Metadata table	Purpose
StateMetadata	State labels, such as Active / Inactive
StatusMetadata	Status Reason labels within each state
OptionsetMetadata	Labels for local choice columns
GlobalOptionsetMetadata	Labels for shared global choices
TargetMetadata	Lookup targets: related table and key column

Choice metadata translates numeric codes into business labels.

Join status labels on both State and Status. Scope label joins by table and language.

Include the metadata your reporting needs alongside the business tables.

Account state and status travel together

State describes the lifecycle. Status Reason explains the state.

statecode	State	statuscode	Status Reason
0	Active	1	Active
1	Inactive	2	Inactive

StatusMetadata must match both State and Status.

Fourth Coffee: Inactive / Inactive after the source change reaches Fabric.

Metadata: EntityName = 'account', language 1033, IsUserLocalizedLabel = 0

Fabric insights back in business apps

VIRTUAL TABLES

Expose Fabric data in Dataverse apps (preview).

The OneLake route is read-only.

WRITE-BACK

Update native Dataverse records through a separate pipeline or API integration.

EXAMPLE: CUSTOMER HEALTH SCORE

Calculate the score in Fabric.
Display it in an app via a virtual table, or save it to a native Account column using a separate pipeline or API.

- SETUP: In Power Apps, choose a Fabric workspace, lakehouse and table. Confirm a unique row ID.
- WRITE-BACK: Fabric Copy activity supports Dataverse Upsert. Match IDs or alternate keys.
- FABRIC LINK: Replication into Fabric remains one-way. Plan writes as a separate integration.

Microsoft Learn: Fabric virtual tables

Microsoft Learn: Dataverse Upsert

DEMO 3

Build for Analytics

Some assembly required.

SSMS / Reporting view

BEFORE YOU BUILD

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NOW LAYING

05 · Pass Inspection — Reconcile data, access, and reporting results.

05 · PASS INSPECTION

Validate data, freshness and access

Agree the comparison scope before deciding whether a number is correct.

DATA

Reconcile source keys, row counts and important totals. Check deletes, null lookups and unmatched labels.

FRESHNESS

Record when each comparison was made. Test a known source change through the link and into the report.

ACCESS

Test representative user roles in Fabric and the model. Include both allowed and denied data.

Repeat these checks after schema, identity or model changes. Record the result and the person responsible.

When the linked schema changes

Coordinate the source change, then revalidate downstream dependencies.

Source change	What to expect	What to check
Add a column	Metadata propagation can delay new columns and temporarily delay data sync.	Use Refresh Fabric tables. Confirm the new column before updating the view or model.
Remove a column	Column remains downstream. Future values are NULL; old values can remain on records not updated since deletion.	Review views and reports that still use the column.
Change type or precision	Unsupported for an existing synced column. It can cause persistent sync failures.	Plan removal and re-addition of the affected table. Validate dependencies.

A successful source edit does not prove the link, SQL view and report are ready.

Microsoft Learn: Fabric Link schema and metadata changes

Worked check: an Account KPI does not match

Compare the same IDs, filter, identity and time window across source, link and report.

ACCOUNT	DATAVERSE	LINKED TABLE	REPORTING RESULT
A-101	0 Active	0 Active	Active
A-102	1 Inactive	**0 Active**	**Active**

ACTIVE ACCOUNTS	Source: 1	Report: 2

Investigate before using the KPI. Confirm the source change, check sync and model freshness, then reconcile again.

Illustrative snapshot. Matching counts alone do not prove matching data.

DEMO 4

Validate the Build

Time for the inspection.

SQL 4 CDS / SSMS

05 · PASS INSPECTION

A repeatable inspection checklist

DATA CORRECTNESS

01 Agree scope, filters and identity.

02 Check keys, duplicates and missing references.

03 Resolve labels once per language.

04 Compare currency totals and precision.

05 Check date/time behavior.

FRESHNESS

06 Verify updates and deletes reach analytics.

07 Check source, link and model freshness separately.

ACCESS AND OPERATIONS

08 Test allowed and denied data with representative roles.

09 Monitor sync status, failures and capacity.

10 Assign an owner and a response to failed checks.

BEFORE YOU BUILD

The Build Plan

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INSPECTION PASSED

Understand the source. Choose the route. Build the model. Validate the result.

WHAT TO TAKE BACK

Five things to take back

1 **UNDERSTAND**
Dataverse rows belong to application processes. Learn the context that gives them meaning.

2 **INSPECT**
Use metadata to confirm names, types, lookup targets and keys.

3 **CONNECT**
Choose the query or replication route deliberately. Plan storage, identity and freshness.

4 **MODEL**
Define grain, resolve meaning and build a shared analytical model.

5 **VALIDATE**
Reconcile data and test access. Repeat the checks when the solution changes.

RESOURCES

Resources: Microsoft setup guidance

Documentation for the platform and the two link options.

DATAVERSE

Platform overview

Application data, tables and platform capabilities.

Dataverse overview

LINK TO FABRIC

Setup and configuration

Prerequisites, identity and linked table selection.

Link to Fabric

SYNAPSE LINK

Azure storage and setup

The route using your ADLS Gen2 storage account.

Azure Synapse Link

BOTTOM LABELS ARE CLICKABLE LINKS. OPEN FROM THE SHARED DECK.

RESOURCES

Resources: Fabric sync and latency

References for planning data freshness and answering follow-up questions.

FABRIC LINK FAQ

Capabilities and common questions

Link behavior, migration considerations and low-latency sync differences.

Fabric Link FAQ

LATENCY PLANNING

Plan data freshness

Workload guidance for evaluating how quickly source changes become available.

Plan latency for Link to Fabric

BOTTOM LABELS ARE CLICKABLE LINKS. OPEN FROM THE SHARED DECK.

RESOURCES

Resources: community tools

Discover community tools for metadata, focused queries and SQL exploration.

TABLE STRUCTURE

Metadata Browser

Explore tables, columns, choices and relationships.

Metadata Browser

FOCUSED QUERIES

FetchXML Builder

Choose fields, apply filters and work with linked tables.

FetchXML Builder

SQL EXPLORATION

SQL 4 CDS

Query Dataverse data and update source records.

SQL 4 CDS

XrmToolBox website & Tool Library

BOTTOM LABELS ARE CLICKABLE LINKS. OPEN FROM THE SHARED DECK.

RESOURCES

Resources: Fabric Link operations

Guidance for migration and downstream troubleshooting.

LOW-LATENCY SYNC

Migration and dependencies

Review dependencies before unlinking and relinking.

Low-latency sync

ONELAKE SHORTCUTS

Repair shortcut targets

Create or replace targets for downstream shortcuts.

OneLake shortcut API

SQL ENDPOINT

Refresh SQL visibility

Browser Refresh button or API automation after lakehouse tables are ready.

SQL metadata refresh

BOTTOM LABELS ARE CLICKABLE LINKS. OPEN FROM THE SHARED DECK.

THE BUILD KIT

Take the reusable examples home.
Get the files on GitHub via the QR code on the next slide.

01

DATAVERSE CONNECTION & METADATA TOOLKIT

Parameterized Excel queries for connections, tables, columns and lookup targets.

02

FETCHXML QUERY EXAMPLE

A focused Account query you can load, inspect and adapt in FetchXML Builder.

03

DATAVERSE & FABRIC SQL EXAMPLES

Query the source, inspect linked data and compare the same records.

Change the environment. Reuse the pattern.



THANK YOU

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Slides + demo files